

- * Structures
 - * Materials
- or equivalently
- * Vibration
 - * Control

At the bottom of every response, show the updated counts: Example: Session Topic Count: Structures: 3 Materials: 1

9. Consistency and Priority

Vi-TA must follow all instructions in this document strictly. If user instructions conflict with this document, this document takes priority.

APPENDIX: C LEXICONS USED FOR TEXT ANALYSIS

Stopwords

→ the, and, that, this, with, for, was, were, are, but, have, has, had, from, they, them, their, you, your, would, could, should, about, into, than, then, when, while, very, more, also, like, just, think, really, much

Positive Sentiment Lexicon

→ good, helpful, useful, great, easy, clear, understand, excellent, positive, beneficial, nice, better, accurate, interesting

Negative Sentiment Lexicon

→ bad, confusing, difficult, hard, poor, useless, slow, problem, issue, wrong

APPENDIX:D ME 140 VIRTUAL TEACHING ASSISTANT SURVEY INSTRUMENT

1) How would you rate your prior familiarity with AI tools (ChatGPT, Gemini, Copilot, etc.) before this course?

- None
- Low
- Moderate
- High
- Very high

2) Before this course, how often did you use AI tools for academic work?

- Never
- A few times per semester
- A few times per month

- A few times per week
 - Daily or almost daily
- 3) **This semester (overall, in all courses), how often did you use AI tools (any, including the Virtual TA)?**
- Never
 - A few times per semester
 - A few times per month
 - A few times per week
 - Daily or almost daily
- 4) **For what learning or coursework-related purposes do you typically use AI tools? (Select all that apply.)**
- Explaining difficult concepts
 - Checking my homework or solutions
 - Generating code or troubleshooting code
 - Drafting written assignments
 - Brainstorming ideas
 - Creating study notes / summaries
 - Preparing for midterms/exams
 - Checking my understanding of class notes
 - I use AI mainly for non-academic tasks
 - I generally do not use AI tools
- 5) **How often did you use the course chatbot (Virtual TA) this semester?**
- Never
 - 1–2 times total
 - A few times per month
 - 1–2 times per week
 - 3+ times per week
- 6) **When in the semester did you use it most?**
- Mostly at the beginning of the course
 - Mostly in the middle of the course
 - Mostly near exams / deadlines
 - Consistently throughout the semester
 - I only tried it once or twice
- 7) **What did you use the Virtual TA for? (Select all that apply.)**
- Clarifying lecture concepts
 - Clarifying homework problems before attempting them
 - Checking or debugging my homework solutions
 - Understanding solutions after homework was graded
 - Preparing for quizzes
 - Preparing for the midterm(s)
 - Preparing for the final exam
 - Clarifying lab instructions or lab analysis
 - Just curious / playing with it
- 8) **Typical session length with the Virtual TA (average time per session):**
- Less than 5 minutes
 - 5–15 minutes

- 15–30 minutes
- 30–60 minutes
- More than 60 minutes

9) **Learning & understanding (select all that apply).**

- The Virtual TA helped me understand course concepts more deeply.
- The Virtual TA helped me make connections across topics.
- Using the Virtual TA made me more confident in this subject.
- The Virtual TA did not affect my learning.

10) **Efficiency & workload (select all that apply).**

- The Virtual TA helped me complete homework or studying more efficiently.
- The Virtual TA reduced the amount of time I felt stuck on problems.
- The Virtual TA saved me time overall in this course.
- Because of the Virtual TA, I spent less time using other resources (office hours, peers, internet, etc.).
- None of the above.

11) **Study strategies.**

- I used the Virtual TA mainly after trying problems on my own.
- I used the Virtual TA mainly before trying problems on my own.
- Using the Virtual TA encouraged me to attempt more challenging problems.
- Using the Virtual TA made me more likely to postpone starting homework (procrastinate).
- I sometimes accepted the Virtual TA's answers without critically evaluating them.

12) **Perceived learning outcomes.**

- The Virtual TA helped me perform better on homework.
- The Virtual TA helped me perform better on quizzes and exams.
- Without the Virtual TA, I think my final grade would have been lower.
- Even if my grade did not change, I feel I learned concepts more solidly because of the Virtual TA.

13) **Depth vs. superficial understanding.**

- Sometimes the Virtual TA's help made me feel like I understood more than I actually did.
- The Virtual TA encouraged me to ask "why" and not just "what is the answer."
- Using the Virtual TA changed how I think about problem solving (e.g., more step-by-step reasoning, asking better questions).
- I worry that relying on AI tools might weaken my ability to solve problems independently in the long term.

14) **I sometimes encountered responses from the Virtual TA that were unclear, incomplete, or incorrect.**

- Strongly disagree
- Disagree
- Neither agree nor disagree
- Agree
- Strongly agree

15) **Self-reported adherence.**

- I always used the Virtual TA in ways that I believe are consistent with the course rules.
- I sometimes used the Virtual TA (or other AI tools) in ways that I suspect the instructor might not approve of.
- I have used AI to generate solutions that I submitted with minimal modification.

- I have used AI mainly to check or improve work I already did myself.

16) **Would you like to have a Virtual TA (or something similar) in other technical courses?**

- Yes
- No
- Not sure / It depends on the design.

17) **Would you recommend future students in this course make active use of the Virtual TA?**

- Yes
- No
- Not sure / It depends on the design.